5_popquiz electric field , dipoles, capacitor

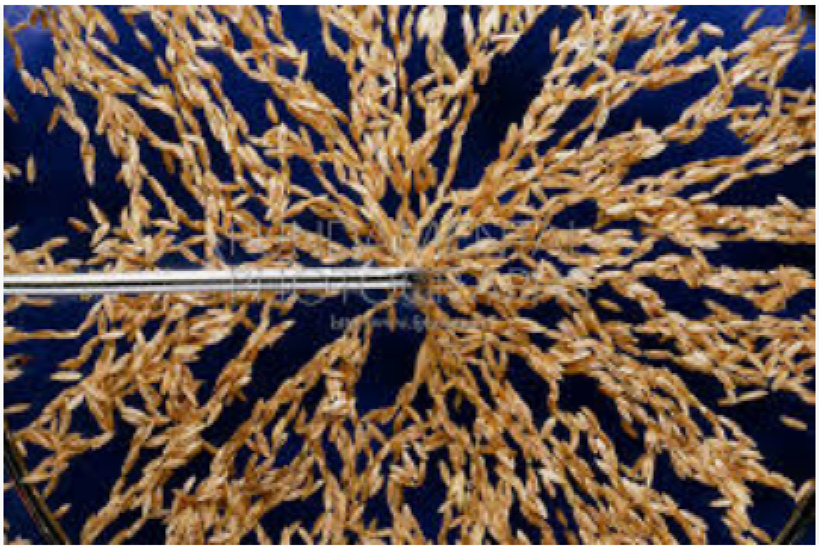
This is a preview of the published version of the quiz

Started: Mar 15 at 3:10pm

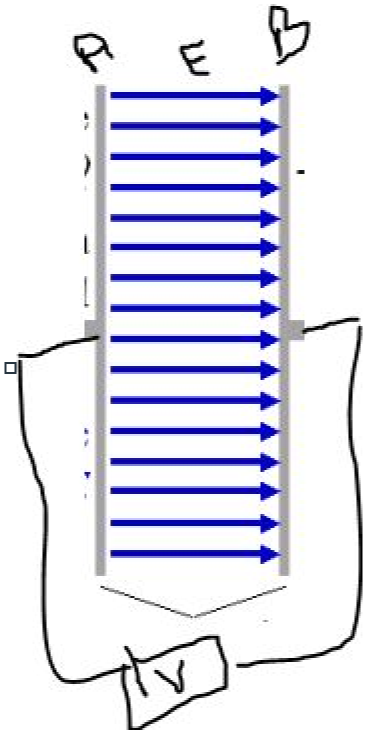
Quiz Instructions

Extra points today !!!   YEY

▫Electric field of a single charge. seeds in oil.



Question 1 15 pts



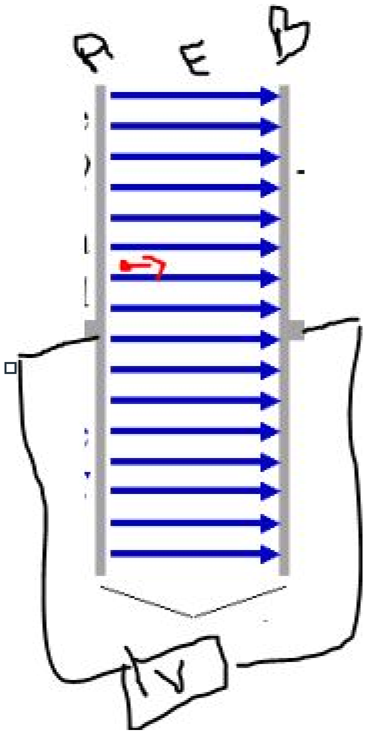
2 plates (conductors) connected to a power supply

This is a charged capacitor. The blue arrows represent the electric field. Which statement is true

uniform means constant, every where same magnitude and direction.

- plate A is negative, E is not uniform
- plate A is positive, E is uniform
- plate A is positive, E is not uniform
- plate A is negative, E is uniform

Question 2 15 pts



A electron is placed between the plates A and B but very close to A. It is given an initial positive velocity (an initial velocity Vo @ right like for the HW) . What will happen? (after it is released). Its tricky ! THINK

hint: the electrons are acted by an electric force. Which way.

- The electron will speed up along horizontal @ right
- The electron will be defected down (don't ask me why)
- The electrons will slow down along horizontal @ right
- The electron will keep the same velocity along horizontal Vo @ right

Question 3 17 pts

A proton is placed in an electric field of intensity 800 N/C. What are the magnitude and direction of the acceleration of the proton due to this field? (e = 1.60 × 10-19 C, mproton = 1.67 × 10-27 kg)

- 5 10^9 m/s/s in the direction of the electric field
- 7.66 10^10 m/s/s opposite to the electric field
- 6 10^9 m/s/s opposite to the electric field
- 7.66 10^10 m/s/s in the direction of the electric field

Question 4 17 pts

A dipole produces an electric field of strength 64 N/C at a distance d (large relative to size of dipole). What is the strength if the distance from the dipole is **halved** so closer ? (still large relative to size of dipole).

hint: z>>d with d the distance between the 2 charges. The E is inversely proportional to the distance .... (see slides)

- 8N/C
- 16N/C
- 256N/C
- 512N/C

Question 5 15 pts

An electric moment vector p (of a dipole) is traced

- is not a vector but a scalar
- from +q to -q
- p is the mass of the dipole times its velocity
- from -q to +q

Question 6 10 pts

The water molecule H2O has a permanent dipole of magnitude p= 6.2 10-30Cm, What is the electric field strength 1 mm from a water molecule at a point **on the dipole axis? That is along the line going through the two charges.**

see slide 71. You have E along the dipole axis at a distance large relative to the size of the dipole.

p = q d   p is the dipole moment. you have the moment of a water molecule.

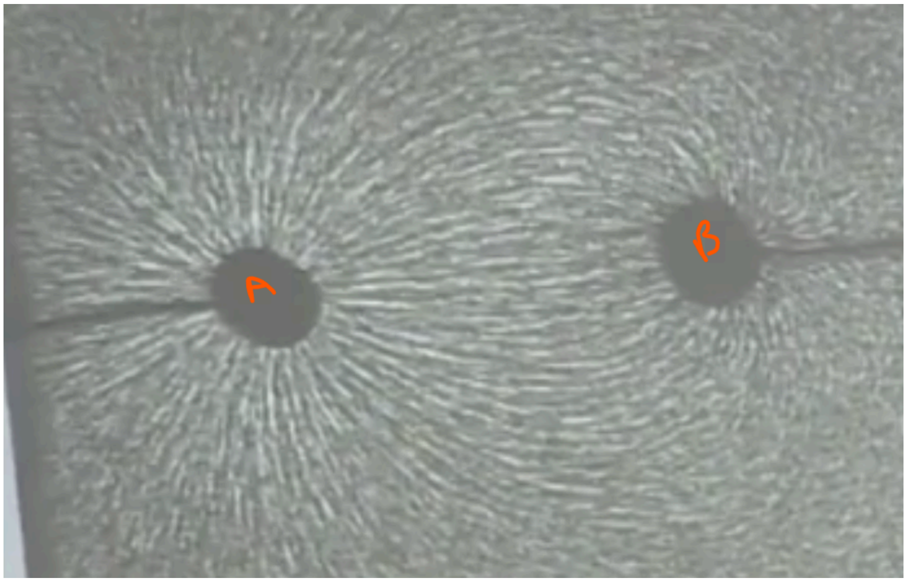
hint: z>>d use the equation K= 9 10^9

- 1.1 10^26 N/C
- 5.5 10^7 N/C
- 1.1 10^-10 N/C

○

1.1 10^35 N/C

Question 7 5 pts



two electrodes A and B are placed in castor coil. Little seeds are sprinkle in the oil.

What is true

○

A has a much larger charge then B in magnitude

○

A and B are not charged

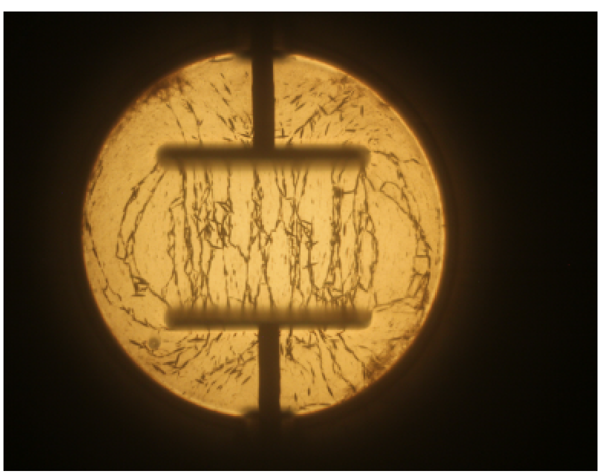
○

A and B have opposite charge

○

A and B are positively charged

Question 8 15 pts

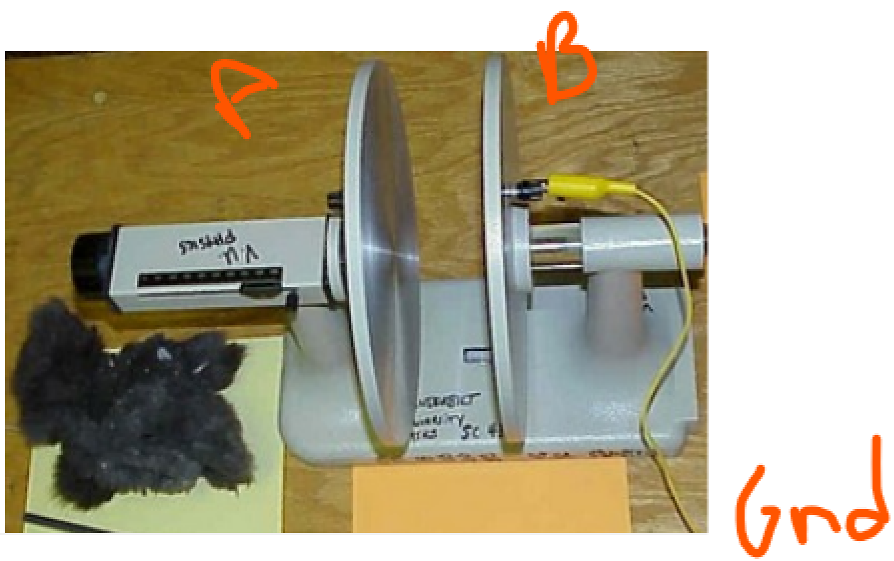


2 electrodes (plates) inside castor oil. With seeds inside. What is true

(slide 78 and lecture)

- The electric field is non uniform between the plates.
- between the plates, the E is uniform
- between the plates, E = 0
- The magnitude of E depends on the distance of the plate

Question 9 15 pts



To get extra credits you build a capacitor. 2 conductors separated by a small distance. You rub a rubber stick (behaves like a balloon) with fur and transfer the charge to A.

B is connected to Gnd (reservoir for charges)

what is true

- A is negative and B is positive
- A is positive and B is negative
- A is negative and B stays neutral
- I don't pay attention in class. oops

Not saved Submit Quiz